

Udit Jain

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SUMMARY

Results-driven Software Development Engineer and M.Tech (CSE) graduate from NIT Warangal (Graduating 2026) with experience at Intel, Cisco and C-DOT. Proficient in **Java, C++, Spring Boot, and ReactJS**, with expertise in **Data Structures, Algorithms, Operating Systems, and Object-Oriented Design**; passionate about building **scalable, reliable, and maintainable systems** through **innovation and collaboration**.

PROFESSIONAL EXPERIENCE

Intel Corporation - Bengaluru, Software Engineer Intern Jul 2025 – May 2026

- Re-engineered the **platform's query pipelines** to support complex, multi-conditional data discovery; built an automated, real-time sorting mechanism that accelerated cross-functional teams' access to the latest platform rollouts and high-volume data streams.
- Developed an automated metrics **aggregation engine from scratch**, consolidating complex product statistics into a single, high-precision reporting system; integrated strict backend calculation logic that eliminated manual data extraction overhead for multiple global Business Units.
- Led a platform-wide **UI/UX optimization** to abstract complex legacy alphanumeric identifiers into intuitive, user-friendly aliases; successfully removed operational friction, minimized data navigation errors, and saved significant development hours for internal engineering teams.

CISCO Systems - Bengaluru, Software Engineer Intern May 2025 - Jul 2025

- Constructed an **AI-driven automation system** using **LangChain and LLMs** to automate software patch deployment, simulation, and report generation for router hardware systems, reducing manual intervention by **70%**.
- Devised the overall **system design** to seamlessly integrate multiple components for **Jenkins** execution, patch loading, and **log** analysis.
- Programmed a scalable **Webex chatbot** with **four** intelligent agents, accelerating **regression testing** and reducing debugging time by **38%**.
- Applied **multithreading** to create **parallel agent instances**, enabling concurrent automation tasks and refining overall system throughput by **25%**.
- Configured modular scripts for **CI/CD integration** and made autonomous result tracking, ensuring real-time visibility of execution status and reports.

Centre for Development of Telematics (C-DOT) - Delhi, Software Engineer Jan 2023 - Jul 2024

- Led **full-stack** feature development by applying **design patterns** and scalable **architecture principles** of the NCCS platform homepage using **Spring Boot, ReactJS, MySQL, and Thymeleaf**, raising load speed by **35%** and enhancing accessibility for internal government users.
- Implemented modular **RESTful APIs** using **Object-Oriented Programming (OOP)** principles and **caching** strategies to enhance backend efficiency.
- Collaborated with a **12-member cross functional** team to ship complex production-ready features with near-zero post-deployment issues.
- Utilized **Git** for branch management, version control, and peer reviews to ensure seamless integration and timely delivery.
- Contributed to hardware automation in the POTP project by validating **ROADM5 configurations** and debugging **30+ processor cards (82xx, 85xx, T1022)**, elevating system uptime and trimming setup time by **25%**.

Infosys Technologies Private Limited - Bengaluru, Digital Specialist Engineer Jul 2022 - Dec 2022

- Designed and deployed a **scalable e-commerce web platform** using the **MERN stack**, serving **10,000+ daily users** for a major client.
- Enhanced system performance by redesigning the **ReactJS** frontend and optimizing **Node.js** and **MongoDB** pipelines.
- Conducted comprehensive **API** testing using **Postman**, achieving a **25%** faster server response and smoother navigation.
- Coordinated with backend and DevOps teams to ensure **reliable API integrations**, version control via Git, and CI/CD automation for faster releases.

SKILLS

Languages	- Core Java, Java 8, JavaScript, Python	Databases	- SQL (MySQL), NoSQL (MongoDB)
Frameworks & Libraries	- Spring Boot, ReactJS, LangChain	Operating System	- Linux, Windows, macOS
Tools and Technologies	- Git, Jenkins, Postman, REST APIs	CS Foundations	- System Design, OOPs, Data Structures, Algorithms

PROJECTS

TrustX-Defect, Explainable Software Defect Prediction Pipeline

Tech Stack: Python, XGBoost/Random Forest, SHAP, Scikit-Learn, Pandas, Matplotlib

- Engineered an **advanced Explainable AI (XAI) pipeline** across 12 cross-project transfer configurations using PROMISE benchmarking datasets to evaluate model generalization under severe distribution shifts.
- Introduced 3 novel behavior-grounded metrics—**Gradient-of-Loss Reliability (GLR)**, **Expected Calibration Error (ECE)**, and **-Perturbation Hit@k**—synthesizing them into a unified **Reliability Score (RS)** framework to formally evaluate XAI trustworthiness.
- Formulated a reliability-aware optimization strategy using train-only preprocessing, probability calibration, and threshold tuning; significantly stabilized prediction precision and achieved near-perfect feature causal relevance (0.9998 Hit@k) in cross-project prediction.

NeuroPredict, Parkinson's Disease Detector

Tech Stack: Python, CatBoost, XGBoost, Scikit-Learn, NumPy, Pandas, Matplotlib

- Independently architected a machine learning model to detect Parkinson's disease from vocal biomarkers using **22+ extracted acoustic features** (jitter, shimmer, NHR).
- Achieved **96.6% accuracy** and **91.4% MCC**, outperforming baseline models; optimized via **SMOTE balancing** and **GridSearchCV** for precision >94%.
- Built a **fully automated ML pipeline** with cross-validation and result visualization for reproducible outcomes and **packaged the project via a virtual environment for local deployment and quality assurance**.

CERTIFICATIONS

Amazon ML Summer School 2025

- Completed intensive training on **machine learning, deep learning, and production-grade ML systems**.
- Gained hands-on exposure to **supervised and unsupervised learning, model evaluation, model optimization, and large-scale ML applications**.
- Strengthened foundations in **AI model design, data preprocessing, and performance tuning** for real-world deployment scenarios at Amazon.

EDUCATION

M.Tech. - Computer Science and Engineering <i>National Institute of Technology, Warangal</i>	<i>Aug 2024 - May 2026</i> CGPA - 8.33
B.Tech. - Computer Science and Engineering <i>Maharaja Agrasen Institute of Technology, New Delhi</i>	<i>Aug 2018 - Jul 2022</i> CGPA - 9.06